

Natural Language Processing

1. Introduction and logistics

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Notice

Credits

The present slides are largely an adaptation of existing material, including slides by [Jurafsky & Martin](#), [Manning](#), [Potts & MacCartney](#), [Black & Mortensen](#), and others, to whom I am very grateful.

Downloading and sharing

A copy of these slides can be downloaded from [virtuale](#) and stored for personal use only. Please do not redistribute.

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Welcome to Natural Language Processing

What is NLP?

Automating the analysis, generation, and acquisition of human (“natural”) language

- Analysis (or “understanding” or “processing” ...)
- Generation (recently: “Generative AI”)
- Acquisition

Why NLP?

- Answer questions using the Web or all sorts of documents
- Translate documents from one language to another
- Do library research; summarize
- Estimate public opinion; read everything and make predictions
- Follow directions given by any user
- Fix your spelling or grammar
- Manage messages intelligently; flag fake news or hate speech
- Write poems or novels
- Listen and give advice
- Write code
- Interactively help people learn, help disabled people, help refugees/disaster victims, document or reinvigorate indigenous languages, ...

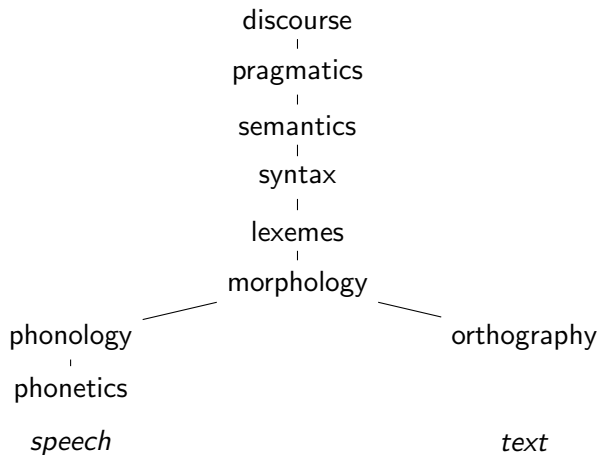
What is NLP? (more detail)

Automating the analysis, generation, and acquisition of human (“natural”) language

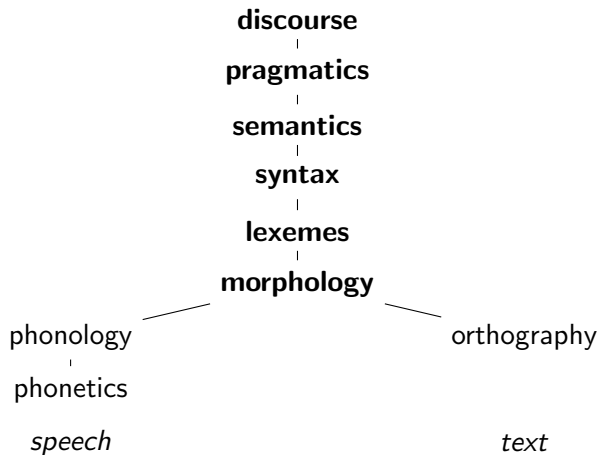
- Analysis (or “understanding” or “processing” ...): input is language, output is some **representation** that supports useful action
- Generation: input is that **representation**, output is language
- Acquisition: obtaining the **representation** and necessary algorithms, from knowledge and data

Representation?

Levels of Linguistic Representation



Let's Examine Some of the Levels



Morphology

Analysis of words into meaningful components

Spectrum of complexity across languages

- *Analytic* or *Isolating* languages (e.g., English, Chinese)
- *Synthetic* languages (e.g., Turkish, Finnish, Hebrew)

Examples

- evlerindekilerin
- 没人需要空格来分隔单词
- unfriend, Obamacare, Bill's

Lexical Analysis

- Normalize and disambiguate words
- Words with multiple meanings: bank, mean
 - Extra challenge: domain-specific meanings
- Multi-word expressions
 - make ... decision, take out, make up, ...
- For English, part-of-speech tagging is one very common kind of lexical analysis

Syntax

- Transform a sequence of symbols into a hierarchical or compositional structure
- Closely related to linguistic theories about what makes some sentences well-formed and others not. For example:
 - I want a flight to Tokyo
 - I want to fly to Tokyo
 - I found a flight to Tokyo
 - I found to fly to Tokyo

Ambiguities

Prepositional phrase attachment ambiguity

San Jose cops kill man with knife

Text

Paper

Translate

Listen

Close

San Jose cops kill man with knife

Ex-college football player, 23, shot 9 times allegedly charged police at fiancée's home

By Hamed Aleaziz
and Vivian Ho

Thursday:

Police officials said two

that, it turned out, had
been made by Watkins

shortly after she called a
suicide intervention
hotline in hopes of get-
ting Watkins medical
help.

Watkins' 911 call came
in at 5:01 p.m., said Sgt.

ed help from police."

She said Watkins was
on the sidewalk in front
of the home when two
officers got there. He was
holding a knife with a
4-inch blade and ran

ing for their safety and
defense of their life, fired
at the suspect."

On the police radio,
one officer said, "We have
a male with a knife. He's
walking toward us."

Examples taken from <http://web.stanford.edu/class/cs224n/>

Ambiguities

Prepositional phrase attachment ambiguity

The image is a screenshot of a BBC News website. At the top, there is a navigation bar with the BBC logo, a 'Sign in' button, and links for News, Sport, Reel, Worklife, Travel, Future, and Mo. Below this is a large red banner with the word 'NEWS' in white. Under the banner is another navigation bar with links for Home, US Election, Coronavirus, Video, World, UK, Business, Tech, Science, and Stor. The main content area features the article title 'Scientists count whales from space' in large black font. Below the title, it says 'By Jonathan Amos' and 'BBC Science Correspondent'. The date '1 November 2018' is shown on the left, and social media sharing icons (Facebook, Messenger, Twitter, Email, and a green 'Share' button) are on the right.

BBC Sign in News Sport Reel Worklife Travel Future Mo

NEWS

Home US Election Coronavirus Video World UK Business Tech Science Stor

Science & Environment

Scientists count whales from space

By Jonathan Amos
BBC Science Correspondent

🕒 1 November 2018

f Messenger Twitter Email Share

Examples taken from <http://web.stanford.edu/class/cs224n/>

Ambiguities

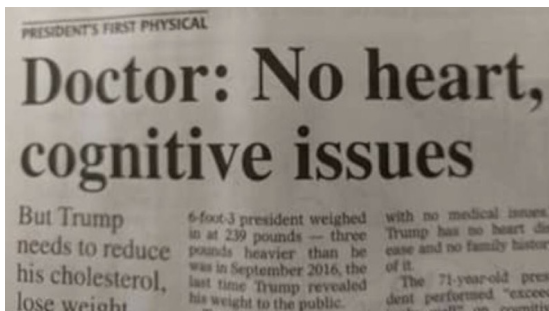
Verb phrase attachment ambiguity



Examples taken from <http://web.stanford.edu/class/cs224n/>

Ambiguities

Coordination scope ambiguity



Examples taken from <http://web.stanford.edu/class/cs224n/>

Ambiguities Explode Combinatorially

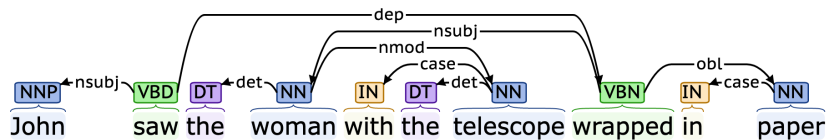
What possible syntactic analyses?

- John saw the woman with the telescope wrapped in paper

Ambiguities Explode Combinatorially

What possible syntactic analyses?

- John saw the woman with the telescope wrapped in paper



Credits: Stanford Core NLP online dependency parser <https://corenlp.run/>

Semantics

In this country a woman gives birth every 15 minutes.

Our job is to find that woman, and stop her. (Groucho Marx)

- Mapping of natural language sentences into domain representations
 - Robot command, database query, formal logic expression, ...
- Scope ambiguities
 - A seat is available to every customer
 - A web site is available to every customer
- Going beyond specific domains is a goal of AI

Pragmatics

- Any *non-local* meaning phenomena
 - “Can you pass the salt?”
 - “Are you 18?” “Yes, I’m 25.”

Discourse

I am happy to join with you today in what will go down in history as the greatest demonstration for freedom in the history of our nation.

Five score years ago, a great American, in whose symbolic shadow we stand today, signed the Emancipation Proclamation. This momentous decree came as a great beacon light of hope to millions of Negro slaves who had been seared in the flames of withering injustice. It came as a joyous daybreak to end the long night of their captivity.

But one hundred years later, the Negro still is not free. One hundred years later, the life of the Negro is still sadly crippled by the manacles of segregation and the chains of discrimination.

(Martin Luther King, Jr.)

- Structures and effects in related sequences of sentences
- Texts, dialogues, multi-party conversations

Why It's Hard

- Input is likely to be noisy
- Linguistic representations are **theorized constructs**; we cannot observe them directly
- Difficult to obtain training data for each aspects
- The mappings between levels are extremely complex
- Appropriateness of a representation depends on the application

Ambiguity

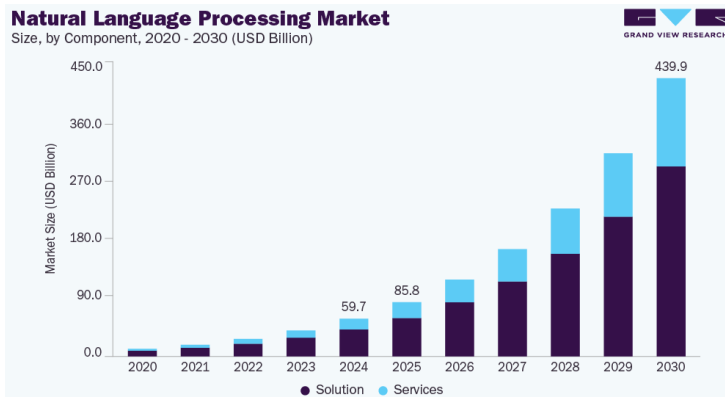
- Each string may have many possible interpretations at every level
- Correct resolution of the ambiguity depends on the **intended meaning**, which is often inferable from context
- People are good at linguistic ambiguity resolution
- Computer not so, as it often requires *understanding*

Complexity of Linguistic Representations

- **Richness:** there are many ways to express the same meaning, and immeasurably many meanings to express. Lots of words/phrases
- Each level interacts with the others
- Tremendous diversity in human languages
 - Languages express the same kind of meaning in different ways
 - Some languages express some meanings more readily/often

NLP market size

NLP is very hard, but also very rewarding



Grand View Research

Aims of this course

This course is meant to

- provide you with the essential NLP background to help you navigate this fast-expanding, challenging, exciting field
- introduce methodologies, basic tools and technological advances of the last years
- help you grow competence and skills required for building simple NLP systems
- discuss perspectives on the most recent evolutions of the discipline

Many Related Disciplines

- **Natural Language Processing** is a branch of AI focused on the technology of processing language (includes NLU, NLG)
- **Computational Linguistics** is focused on the technology to support/implement linguistics; e.g., study of language change over time
- **Information Retrieval** is about obtaining information system resources that are relevant to an information need from a collection of those resources; e.g., Web search engines
- **Text Mining & Big Data Analytics** are subtypes of Data Mining, focused on search and retrieval of information from large bodies of text; e.g., information discovery; biomedical text mining supporting clinical needs

In This Course...

- Syntactic and lexical analysis
 - edit distance, part-of-speech tagging
- Semantics
 - document encoding, language modeling, distributed semantics, embeddings
- Architectures and Tools
 - basic text processing tools
 - basic neural architectures: RNNs, CNNs, sequence-to-sequence
 - advanced attention-based and transformer-based architectures
- Large Language Models
 - encoder vs decoder models, fine-tuning vs in-context learning, prompting, pre-training, model alignment, evaluation methods
- Applications
 - analytical: text classification, sentiment analysis, information retrieval, question answering, argument mining, ...
 - generative: question-answering with RAG, machine translation

Course Organization and Logistics

Teaching Staff and Audience

Teaching staff

- Instructor: Paolo Torroni
- Teaching assistants: Federico Ruggeri & Eleonora Mancini
- Other members of the [Language Technologies Lab](#)

Intended audience

- 2nd-year students of the Masters' Degree in AI
- Anyone with an interest in NLP and who is familiar with
 - Statistical and mathematical methods for AI
 - Machine learning and deep learning concepts, methodologies and tools
 - The Python programming language

Learning Resources

- Course slides

- Textbook

- [Speech and Language Processing](#), by Dan Jurafsky and James H. Martin. 3rd Ed. online draft.

- Selected articles, blogs, and other online resources

- Further reference textbooks

- [Natural Language Processing](#), J Eisenstein. Online draft.
 - [Natural Language Processing with Python](#), S Bird et al. O'Reilly
 - [Natural Language Processing with Transformers](#), L Tunstall, O'Reilly.
 - [Natural Language Processing with PyTorch](#), D Rao et al. O'Reilly.
 - [Deep Learning: Foundations and Concepts](#), CM Bishop et al. Springer.

- Freely available software

- [NLTK](#), [Colab](#), [PyTorch](#), [Huggingface](#), [LangChain](#)



Assessment

- Two **assignments** (6 points each)
 - For each assignment delivered by January 9, 2025: + 0,5 points
 - Exceptionally: + 0,5 points if work done is particularly clever
 - To be done **in groups of 3/4**
 - Changing groups between assignments is permitted, even welcome
 - Piece of advice: start looking for team-mates for the assignments today
- A **written exam** on all topics seen in class (18 points)
 - Access to exam only if both assignments submitted by **cut-off dates**
 - Assignments submitted by **9 January** → speed bonus + access to 20 January, 10 February, June, July, September exams
 - Assignments submitted by **31 May** → June, July, September
 - Assignments submitted by **31 August** → September
 - After 31 August → wait for next year's assignments
- Final grade is sum of all points earned
 - Cum laude if sum ≥ 31

Project Work

- **Project Work in NLP (3 cfu).** Some options:
 - work on a challenge, e.g. **SemEval** or **CLEF**
 - explore a foundational aspect
 - explore NLP methods for low-resource languages
 - work on a challenging application/research topic
 - prepare a literature survey on a research topic
 - possibility to follow up PW with work on publication
- Submit project and report in the Virtuale page of PW in NLP (2025/26 edition)
- Exam is project presentation – may be done online
- If you are interested in PW in NLP, contact the teaching staff to agree on a topic before you start working on it

Master Thesis

Thesis opportunities:

- Thesis abroad
 - watch out for grants
 - helpdesk: [International mobility office](#).
- Internship + thesis at a company
 - Many contacts with good companies
 - Check Virtuale page on Internships and theses in NLP
 - Sign up to the AI Internship offers Teams group (if you don't know how to join ask the Programme Coordinator)
- Work with [our group](#) on a research topic

Contacts

- Use **email** not Teams
- Word of advice: **always Cc Federico Ruggeri & Eleonora Mancini** in all communications about the NLP course, and write “NLP course” in the Subject
- That ↑ will significantly reduce your wait for an answer
- Office time by appointment (email)
 - preferably but not necessarily in person
- Contact details of teaching staff on **Virtuale**

Events

- Mid-October to mid-November: 3 hands-on tutorials on NLP methods, libraries and tools, covering text classification, embedding, huggingface transformers, prompting and much more
- End of October: launch of first assignment
- Mid November: launch of second assignment
- This year's course will also feature invited talks, topic highlights, meetings with companies and alumni, and a special track on speech processing

Questions?